GAIT ANALYSIS FOR PARKINSON DETECTION USING MAMBA AND STATE SPACE MODELS

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ABSTRACT

Parkinson's disease (PD) is a progressive neurodegenerative disorder that significantly affects gait and motor control, making gait analysis a valuable non-invasive tool for early detection. In this study, we investigate the use of selective state space models, specifically Mamba, for PD detection using vertical ground reaction force (vGRF) signals. Experiments were conducted on the PhysioNet Gait in Parkinson's Disease dataset, which contains force measurements collected from sensors under the feet of PD patients and healthy controls. The proposed approach is compared with baseline machine learning and deep learning models, including Support Vector Machines (SVM), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU). While recent works using transformer-based architectures have reported classification accuracies above 95% for this task, they often involve substantial computational cost. Mamba offers an alternative sequence modeling paradigm with linear complexity and efficient long-range dependency modeling. Experimental results show that the Mamba model achieves the best overall performance with an accuracy of 88.61%, while also demonstrating faster convergence and smoother training dynamics compared to recurrent architectures. These results highlight the potential of selective state space models as a computationally efficient solution for gait-based Parkinson's disease detection.

1. INTRODUCTION

Parkinson's disease (PD) is a progressive neurodegenerative disorder that affects motor control, balance, and gait. Clinical diagnosis and severity assessment commonly rely on scales such as the Unified Parkinson's Disease Rating Scale (UPDRS) and the Hoehn and Yahr scale, which depend on clinician evaluation and may introduce subjectivity. As a result, objective data-driven approaches based on gait analysis have attracted increasing interest. Vertical ground reaction force (vGRF) signals recorded during walking provide rich time-series information describing locomotion dynamics. Machine learning techniques applied to gait data have shown promising results in distinguishing PD patients from healthy subjects (1). For example, Ertuğrul et al. (2) proposed a method based on Shifted One-Dimensional Local Binary Patterns (Shifted 1D-LBP) to extract discriminative temporal features from gait signals, achieving 88.9% classification accuracy using a multilayer perceptron. More recently, deep learning models such as convolutional neural networks, recurrent neural networks, and hybrid architectures have been applied to automatically learn features from gait signals (3; 4). Transformer-based models have also been explored for this task, achieving classification accuracies above 95% by capturing long-range temporal dependencies in gait sequences (5). Despite their effectiveness, transformer architectures typically require high computational resources due to quadratic attention complexity. Selective state space models, such as *Mamba*, provide an alternative framework for sequence modeling that maintains strong long-range dependency modeling while operating with linear computational complexity. Compared to recurrent networks like LSTM and GRU, these models avoid sequential processing bottlenecks and offer improved training efficiency on long time-series data. In this work, we investigate the use of the selective state space model *Mamba* for Parkinson's disease detection using vGRF signals from the PhysioNet Gait in Parkinson's Disease dataset. The proposed approach is evaluated against traditional machine learning and deep learning baselines, including SVM, LSTM, and GRU, demonstrating competitive performance with improved training stability and computational efficiency.

1.1. Contributions of This Work

The main contributions of this study are threefold. First, we apply the selective state space model Mamba to gait-based Parkinson's disease detection, providing one of the first evaluations of this class of models in this domain. Second, we conduct a systematic comparison with traditional machine learning baselines, such as Support Vector Machines (SVM), and recurrent deep learning architectures including Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), demonstrating that Mamba achieves superior performance with a classification accuracy of 88.61% and improved generalization under comparable training conditions. Finally, we analyze the training dynamics of Mamba and LSTM on the vGRF dataset, showing that Mamba exhibits smoother loss curves, faster convergence, and reduced overfitting, addressing key challenges in modeling long biomedical time-series signals.

2. DATASET

This study utilizes the publicly available Gait in Parkinson's Disease dataset from the PhysioNet PhysioBank repository (6). The dataset comprises gait recordings from 166 subjects: 93 patients diagnosed with idiopathic Parkinson's disease (PD) and 73 age-matched healthy controls (HC). The demographic characteristics of the participants are presented in Table 1.

Table 1. Demographic characteristics of study participants

Characteristic	HC	PD (HY=2)	PD (HY=2.5)	PD (HY=3)
Number of subjects	73	55	28	10
Age (years)	66.3±10.8	64.8±11.2	68.3±9.7	67.5±10.2
Sex (M/F)	40/33	36/19	16/12	6/4
Height (m)	1.71±0.08	1.69±0.07	1.70±0.07	1.68±0.06
Weight (kg)	70.8±13.2	71.2±12.1	72.4±8.9	65.3±7.8

The dataset was collected across three independent studies (7; 8; 9), where participants walked at their self-selected comfortable pace on level ground for approximately two minutes. Gait data were acquired using force-sensitive resistors (Ultraflex Computer Dyno Graphy, Infotronic Inc.) embedded in insoles, with eight sensors positioned under each foot to measure vertical ground reaction force (VGRF). The sensor placement captured force distribution across key plantar regions: heel, mid-foot, and forefoot areas. All measurements were digitized at a sampling frequency of 100 Hz. Each recording includes 18 time-series signals: 16 individual sensor outputs (8 per foot, labeled L1–L8 for left foot and R1–R8 for right foot) plus two aggregate signals representing the total VGRF under each foot, Figure 1.

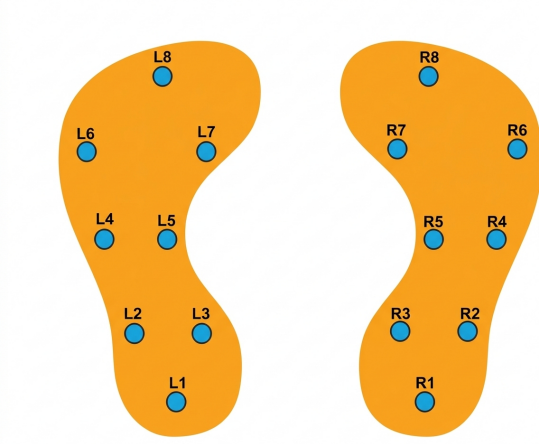


Fig. 1. Position of sensors under each foot insole.

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The VGRF signals are measured in Newtons as a function of time and provide rich spatiotemporal information about gait characteristics. For PD patients, disease severity was assessed using the Hoehn and Yahr (HY) scale, ranging from stage 2 (mild bilateral symptoms) to stage 3 (moderate bilateral disease with postural instability). The distribution of subjects across diagnostic groups and severity levels is shown in Table 2. The dataset includes multiple gait trials per subject recorded under different conditions, including normal walking, dual-task walking, and walking with rhythmic auditory stimulation. Figure 2 illustrates representative total vertical ground reaction force (VGRF) signals for the left and right foot from a healthy control and a patient with Parkinson's disease (PD). The periodic force peaks correspond to consecutive stance phases, while intervals near zero represent swing phases.

Table 2. Distribution of subjects by diagnosis and disease severity

Group	Number of subjects	Percentage	HY stage
Healthy Controls	73	44%	–
PD (Mild)	55	33%	2.0
PD (Medium)	28	17%	2.5
PD (High)	10	6%	3.0
Total	**166**	**100%**	–

Compared to healthy controls, PD subjects exhibit reduced peak force amplitudes, increased stride-to-stride variability, and subtle irregularities in temporal spacing between force peaks. These differences reflect known motor impairments in PD, including bradykinesia, reduced push-off strength, impaired rhythm generation, and altered gait automaticity. The observable differences in force magnitude, periodicity, and variability motivate the extraction of temporal, force-based, and nonlinear dynamical features for downstream classification and severity analysis.

3. PROPOSED APPROACH

In this study i adopted two complementary modeling paradigms for Parkinson's disease detection using vertical ground reaction force (vGRF) signals: Feature-Based Machine Learning Approach and End-to-End Sequence Modeling Approach.

3.1. Feature-Based Machine Learning Approach

In the first modeling paradigm, i adopted a feature engineering based classification pipeline. As an initial step, automated feature extraction was performed using the *TSFRESH* library on the raw vertical ground reaction force (vGRF) signals obtained from the 16 insole sensors. TSFRESH computes a comprehensive subset of time-series descriptors for each sensor and each subject, including basic statistics (mean, median, standard deviation), extrema-based measures, and simple temporal characteristics. The resulting feature matrix contained one row per subject (93 PD patients and 73 healthy controls) and several hundred columns representing engineered gait descriptors derived from all sensors. In addition to the automatically extracted features, a targeted handcrafted

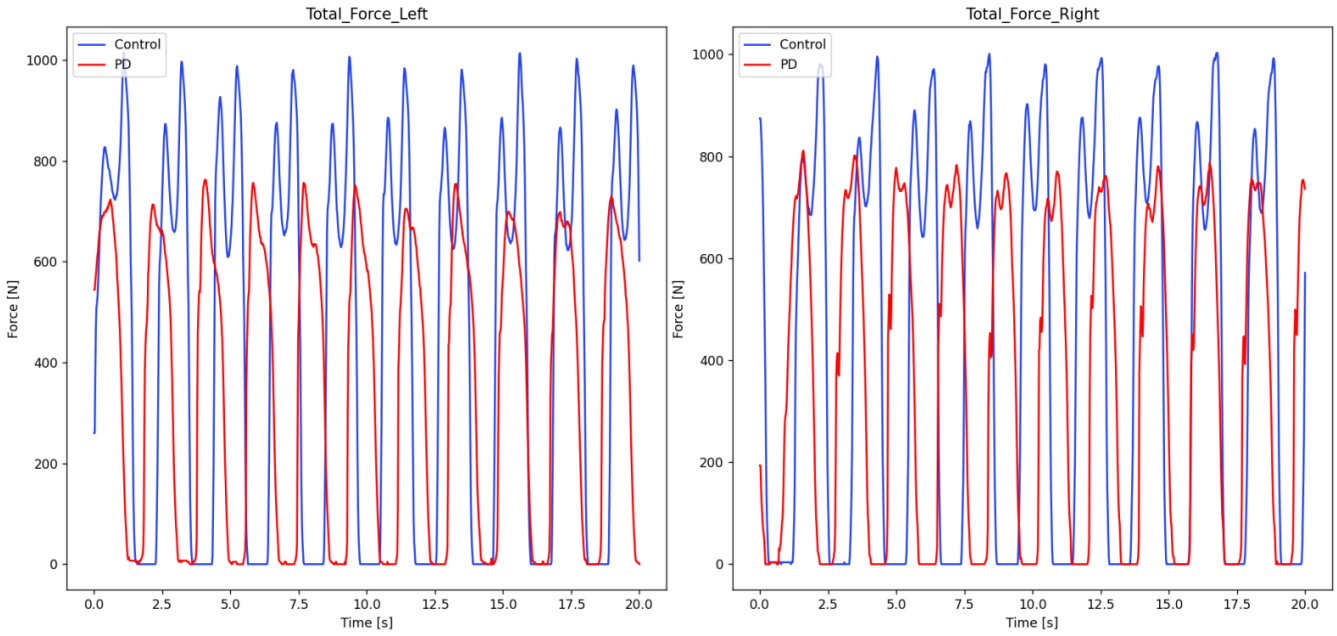


Fig. 2. Representative total vertical ground reaction force (VGRF) signals for the left and right foot in a healthy control (blue) and a Parkinson's disease patient (red).

feature extraction pipeline was developed to capture biomechanical and nonlinear characteristics known to be affected in Parkinson's disease (PD), such as rigidity, impaired rhythm generation, and reduced motor control. The raw force signals were filtered and segmented into individual strides, from which stride-level descriptors were computed. Examples include *L_impulse_cv*, representing the coefficient of variation (CV) of the left foot impulse, and *stance_phase_cv*, quantifying variability in stance duration, both reflecting stride-to-stride inconsistency and postural instability commonly observed in PD. In total 229 features were generated. To analyze redundancy and intrinsic dimensionality, PCA was applied (Figure 3). PC1 and PC2 explain 18.4% and 11.8% of total variance respectively (29.8% combined), and approximately 35 components were required to reach 95%, confirming substantial redundancy in the 229-dimensional feature space. Scientifically, this pattern suggests that gait pathology effects are distributed rather than localized to a single biomechanical signature. Parkinsonian gait alterations appear to arise from many moderate contributors, such as subtle changes in force magnitude, timing variability, asymmetry, and nonlinear dynamics, rather than from one dominant abnormality. To identify the most discriminative features, a ranking strategy combining three criteria was applied: Mutual Information (MI) for nonlinear dependencies, F-score for linear separability, and Cohen's d for effect size. After experimentation with different weight combinations and t-SNE evaluation, the optimal configuration assigned equal weights to Cohen's d and F-score (0.5 each) with MI excluded (0.0), prioritizing effect magnitude and linear discriminative strength. The top nine features selected were:

`L2__maximum`, `L2__absolute_maximum`, `R2__maximum`, `R2__absolute_maximum`, `R6__median`, `L6__median`, `R5__median`, `L8__median`, `L2__standard_deviation`

Notably, all highest-ranked features originated from TSFRESH automated extraction, while handcrafted biomechanical descriptors ranked lower. Increasing the MI weight promoted nonlinear features into the top ranks but consistently produced more overlapping t-SNE clusters, indicating reduced separability. The discriminative capacity of the selected features is illustrated in Figure 4, where they produce the most distinct clustering between PD patients and healthy controls. These features were subsequently used as input to classical classifiers, including linear and RBF-kernel Support Vector Machines (SVM).

3.2. End-to-End Sequence Modeling

In contrast to the feature engineering pipeline, we also investigate an end-to-end sequence modeling approach. In this setting, the raw Vertical Ground Reaction Force (vGRF) signals acquired from the sensors are directly used as input to deep neural architectures, allowing the models to automatically learn temporal patterns and discriminative representations from the data. This approach avoids manual feature de-

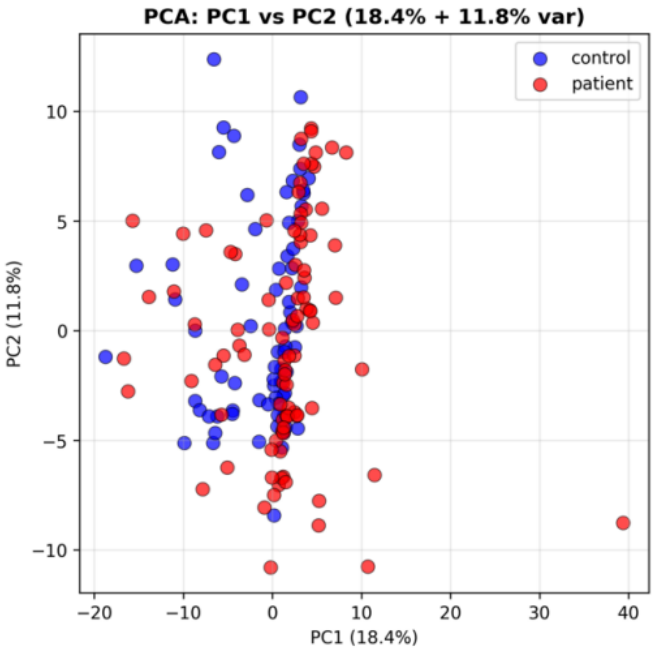


Fig. 3. PCA projection of the 229-dimensional feature space onto the first two principal components (PC1 and PC2). Together, they explain 29.8% of the total variance.

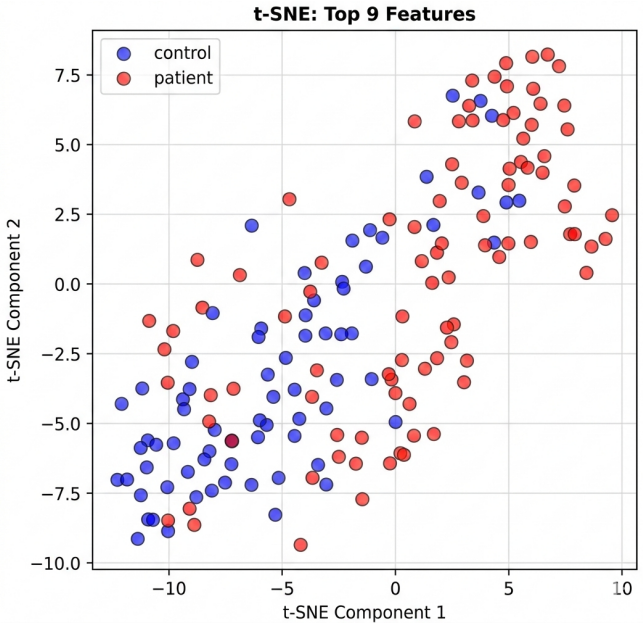


Fig. 4. t-SNE visualization of the top nine ranked features showing improved clustering between Parkinson's patients and healthy controls.

sign and instead relies on the sequence modeling capabilities of recurrent and state-space architectures. In particular, we evaluate three models that are well-suited for temporal data: Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and the recent Mamba architecture, Figure 5. These models are trained to capture temporal dependencies in gait signals and perform classification directly from the sequential sensor measurements.

3.2.1. Data Preprocessing

The same type of preprocessing was applied to all three models. The txt files for patients and controls, containing between 4,000 and 13,000 timesteps, were segmented into chunks of 2,000 timesteps. 2,000 was chosen because good performance was observed in the 1,000-3,000 timestep range, while performance degradation was observed for chunks less than 500 or greater than 3,000. A per-channel z-score normalization was then applied. Eighty percent of the data was used for testing and 20% for validation. The split was done at the subject level to avoid data leakage; this step is particularly important since chunking the original dataset created different data files for the same subject.

3.2.2. Mamba Block

State Space Models (SSMs) are a class of sequence modeling architectures designed to efficiently process long sequential data such as text, audio, or physiological signals. An SSM represents a sequence using a hidden *state* that summarizes the past information observed so far. At each timestep t, the model updates its hidden state h_t based on the current input x_t and the previous state h_{t-1}, and then produces an output y_t. In this formulation, the hidden state acts as a compressed representation of the sequence history, allowing the model to capture temporal dependencies across long sequences. Compared to attention-based models, SSMs have the advantage of linear computational complexity with respect to sequence length, making them particularly suitable for long time-series data. Early structured state space models, such as S4, improved sequence modeling by introducing a structured transition matrix and specialized initialization (HiPPO) to better capture long-range dependencies. However, these models used fixed parameters that were independent of the input, limiting their ability to selectively focus on relevant parts of the sequence. Mamba extends these approaches through the introduction of a *Selective State Space Model*. In contrast to traditional SSMs, Mamba makes parts of the state update mechanism dependent on the input sequence. Specifically, the matrices controlling how the input influences the state are dynamically generated for each timestep, allowing the model to selectively filter relevant information while ignoring

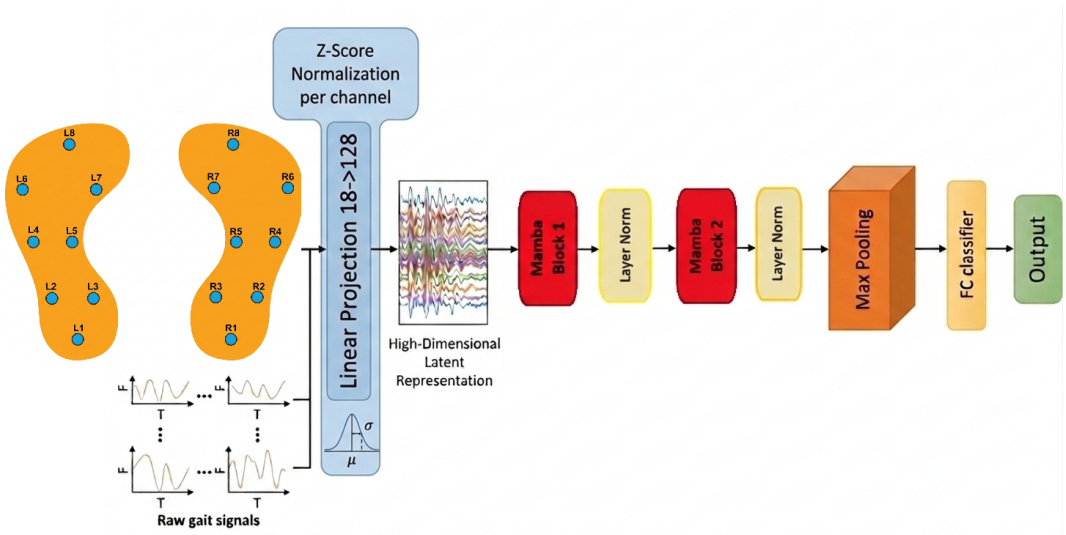


Fig. 5. Architecture of the proposed Mamba-based pipeline used in this study. Raw multi-channel sensor signals from the foot are first normalized using per-channel Z-score normalization and projected into a higher-dimensional latent space through a linear embedding layer. The resulting sequence representation is processed by stacked Mamba blocks (6) with intermediate layer normalization to capture temporal dependencies in the gait dynamics. A max-pooling layer aggregates the learned sequence features, which are then passed to a fully connected classifier to produce the final prediction.

less informative inputs. This selective mechanism enables the model to retain useful long-term context while avoiding the accumulation of irrelevant information. The Mamba architecture is implemented as a neural network block similar to a Transformer layer. As illustrated in Figure 6, the input sequence is first projected into a higher-dimensional representation and processed with a convolutional layer to capture local dependencies between neighboring timesteps. The sequence is then passed through the selective state space module, which performs the state update using the dynamic parameters described above. Finally, the output is projected back to the original dimension and combined with residual connections and normalization layers. Multiple Mamba blocks can be stacked to form deep sequence models capable of modeling complex temporal patterns while maintaining linear computational complexity with respect to sequence length.

4. EXPERIMENTS & RESULTS

A random search was performed to find the hyperparameter values, the results are reported in table 3. The number of channels has also been studied as a parameter. It is reasonable to think that not all sensors are equally discriminating, rFigure 7. For example, in the feature engineering approach, the most discriminating features come from sensors [2,5,6,8], respectively [2,5,6,8,10,13,14,16] for the left and right foot.

Table 3. Training and architectural hyperparameters for each sequence model.

Parameter	LSTM	GRU	Mamba
Channels	[2,5,6,8,10,13,14,16,17,18]	all	all
Batch size	32	32	64
Hidden dim	64	96	128
Layers	2	3	2
Dropout	0.3	0.3	—
Learning rate	5e-3	2.5e-4	5e-5
Optimizer	Adam	Adam	Adam
Loss	BCE	BCE	CE (w. 1.1/1.0)
ES patience	10	10	10
LR sched. patience	5	5	5
d_{state}	—	—	32
d_{conv}	—	—	32

4.1. Results

Table 4 summarizes the overall classification performance of the evaluated models on the gait dataset. Among all approaches, the Mamba model achieves the best overall performance with an accuracy of 88.6%, outperforming both recurrent neural networks and traditional machine learning baselines. The LSTM model follows closely with an accuracy of 87.2%, while the GRU reaches 84.4%. Traditional machine learning approaches such as the RBF SVM and Linear SVM show lower performance, achieving accuracies of 80.6% and 79.4%, respectively. The superiority of the Mamba architecture is also reflected in the precision, and F1-score metrics. In particular, Mamba achieves the highest precision (0.897) and the highest overall F1-score (0.885), indicating strong balance between sensitivity and specificity

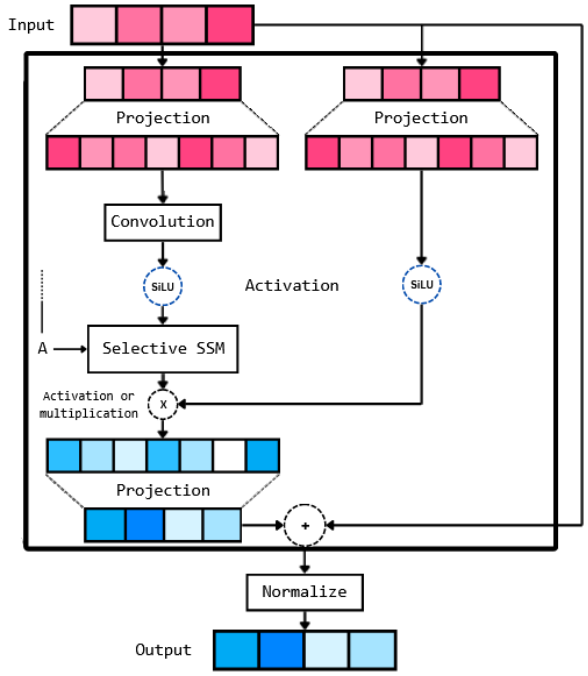


Fig. 6. Architecture of a Mamba block. The input sequence is projected and processed by a convolution layer before passing through the Selective State Space Model, which dynamically updates the hidden state based on the input. The block can be stacked to form deep sequence models.

when distinguishing between Parkinson's disease and control subjects.

Table 4. Overall Classification Performance of Different Models

Model	Accuracy	Precision	Recall	F1-Score
Mamba	**0.886**	0.897	0.886	0.885
LSTM	0.872	0.875	0.872	0.872
GRU	0.844	0.846	0.844	0.843
RBF SVM	0.806	0.803	0.803	0.803
Linear SVM	0.794	0.791	0.791	0.791

Figure 8 provides a detailed comparison of the models across class-specific metrics. The results show that deep learning models generally outperform traditional SVM approaches. Mamba demonstrates the strongest overall performance, LSTM also perform competitively but show slightly lower scores across most metrics. In contrast, the SVM-based models exhibit weaker recall and F1-scores, suggesting that classical models struggle to fully capture the complex temporal dynamics present in gait signals. To further analyze model behavior during training, the learning dynamics of Mamba and LSTM were compared using their training and validation loss curves. Figure 9 illustrates the training dynamics of both architectures. The Mamba model exhibits a smooth and consistent decrease in both training and validation loss, indicating stable learning and effective generalization. The gap between the training and validation curves remains relatively small, suggesting that the model is able to capture relevant patterns without significant overfitting. In contrast, the LSTM model shows more irregular validation loss fluctuations during training. Although the training loss continues to decrease, the validation loss becomes unstable after approximately epoch 30, indicating the onset of overfitting. This behavior highlights one of the advantages of state-space based architectures such as Mamba, which are designed to model long-range temporal dependencies more efficiently while maintaining stable training dynamics. Overall, these results demonstrate that the Mamba architecture provides a robust and effective approach for modeling gait time-series data, achieving the highest classification performance while also exhibiting improved training stability compared to recurrent neural network baselines.

5. DISCUSSION

The results demonstrate that selective state space models are effective for modeling gait time-series data in Parkinson's disease detection. Among the evaluated models, the Mamba architecture achieved the best overall performance and showed more stable training behavior compared to recurrent neural networks such as LSTM and GRU. This suggests that models designed to efficiently capture long-range temporal dependencies are well suited for vertical ground reaction force signals, which contain complex temporal patterns related to gait dynamics. The comparison between modeling approaches also highlights the advantage of end-to-end sequence models over feature-based pipelines. While engineered features captured relevant statistical properties of the gait signals, deep learning models were better able to learn discriminative temporal representations directly from the raw data. These findings are consistent with previous studies showing improved performance of deep sequence models for gait analysis. However, while transformer-based models reported in the literature can achieve even higher classification accuracy, they typically rely on computationally expensive attention mechanisms. In contrast, the Mamba architecture provides a more computationally efficient alternative with linear complexity while still achieving competitive performance. The biggest limitation is due to the relatively small and unbalanced dataset, with 30% of the data for the control class and 70% for the Parkinson's class. In fact, chunk-based segmentation was necessary for training sequence models. Single signal modality may limit generalization, suggesting that larger multimodal datasets could further improve performance.

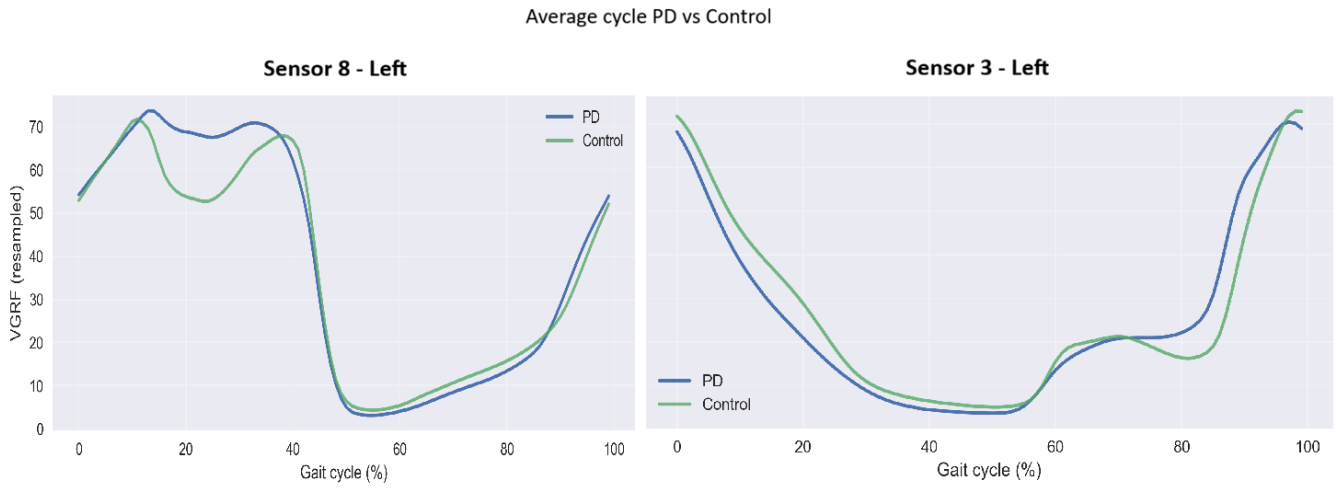


Fig. 7. To visually compare gait dynamics between Parkinson's disease (PD) patients and healthy controls, average gait cycle plots were generated for each sensor channel. For every sensor, two plots are produced corresponding to the left foot and right foot. Each plot displays the mean normalized gait cycle computed separately for the two subject groups. The gait signals are first segmented into individual gait cycles and temporally normalized so that each cycle is represented over a fixed number of time points. The cycles belonging to the same group are then averaged to obtain a representative waveform for that sensor, enabling direct visual comparison of pressure dynamics throughout the gait cycle.

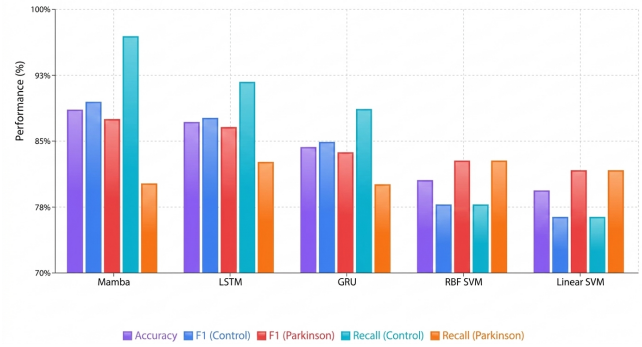


Fig. 8. Performance comparison of the evaluated models across multiple metrics, including accuracy, F1-score, and recall for both control and Parkinson classes.

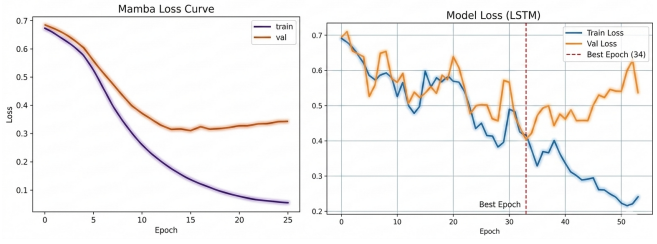


Fig. 9. Training and validation loss curves for Mamba and LSTM models. Mamba shows smoother convergence and more stable validation behavior compared to LSTM.

6. CONCLUSION

These findings demonstrate that selective state space models represent a promising approach for modeling biomedical time-series data such as gait signals. By combining efficient sequence processing with strong temporal representation learning, Mamba provides a computationally efficient alternative to both recurrent and attention-based architectures. Future work could extend this approach by incorporating additional sensor modalities and evaluating the model on larger and more diverse clinical datasets. Such developments could further improve the reliability of automated gait analysis systems and contribute to the development of non-invasive tools for early Parkinson's disease detection and monitoring.

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